ON TOPOLOGICAL PROPERTIES OF MIDDLE GRAPH OF BENZENOID PLANAR OCTAHEDRON NETWORKS

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IN
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A thesis submitted in partial fulfillment of the requirements for the degree of

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In the Name of
Holy Allah
the Most Beneficent and
the Most Merciful

Dedicated

to

My Parents, Teachers and my Family

DECLARATION

The work reported in this thesis was carried out by me under the supervision of **Dr. Shah jahan and Dr. Zaheer Ahmad** Institute of Mathematics, Khawaja Fareed university of Engineering and Information Technology, Rahim Yar Khan, Pakistan.

I hereby declare that the title of the thesis **"On Topological Properties of Middle Graph of Benzenoid Planar Octahedron Networks"** and the content of the thesis are the product of my own research and no part has been copied from any published source (except the references, standard mathematical or genetic models / equations / formulas / protocols etc. I further declare that this work has not been submitted for an award of any other degree/diploma.

At any time if my statement is found to be incorrect even after my graduation the university has the right to withdraw my MS degree.

Name: MUHAMMAD SAJJAD

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Certificate of Approval

This is to certify that the research work presented in this thesis, entitled "**On Topological Properties of Middle Graph of Benzenoid Planar Octahedron Networks**" was conducted by **Muhammad Sajjad (Math211501005)** under the supervision of **Dr. Shah jahan** and **Dr. Zaheer Ahmad**

This thesis is submitted to the institute of Mathematics, Khawaja Fareed University of Engineering and Information Technology, Rahim Yar Khan in partial fulfillment of the requirements for the degree of Master of Science in field of Mathematics.

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MUHAMMAD SAJJAD

ABSTRACT

The analysis of the fundamental structural characteristics of middle graph benzenoid planar octahedron networks, such as the configuration of benzene rings, the existence of connecting bridges, and the overall planarity of the network, is the first step of the research. These structures are investigated and characterized using a variety of computational methods, including graph theory and molecular modelling. In this thesis, we examine the topological characteristics of the middle graph in the framework of a network of benzenoid planar octahedron. A crucial component that captures the connection and structural configurations of molecules or networks is the middle graph. In this article, we give a thorough introduction to benzenoid planar octahedron networks, emphasizing their importance in organic chemistry and materials science. We investigate their distinctive geometrical characteristics, including the configuration of the carbon atoms and the presence of six-membered rings. The benzenoid planar octahedron network's middle graph is defined and examined. We study its building process, concentrating on how it depicts the network's central region while retaining its key topological characteristics. We examine the connections between the middle graph and the entire network, assessing their dependency and determining what it means for the features of the network. We illustrate the dependability and correctness of our suggested approaches for characterizing the topological characteristics of the middle graph through computer studies and comparisons with existing experimental data. These molecular descriptors are frequently employed to describe animate effects based on chemical structures in the research of quantitative structure-activity relationships (QSARS). Based on vertex degree, the substance diagram hypothesis established the idea of topological indices. After converting a network into a line graph, we use formulas for degree-based topological indices of line graphs of carboxymethyl guar to determine the degree of each vertex. Next, we study topological indices like the Randic Index, First, Second, and Third Zagreb Index, Geometric arithmetic index (GA), First and second and Zagreb polynomials, Firstly and second hyper Zagreb indices, Atom Bond Connectivity Index (ABC), First and second hyper Zagreb indices, and forgetting topological indices of the line graph.

Keywords: Middle Graph, Topological indices, Benzenoid planar, Octahedron Graph.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Name**
RI	Randić Index
FI	Forgotten index
SCI	Sum connectivity index
HNI	Hyper Zagreb index
N	Zagreb
G_1	Graph
N_1	First Zagreb index
MBPOH	Middle Graph benzenoid planar Octahedron networks

Chapter 1

INTRODUCTION

In graph theory, a graph consists of two main components: vertices (also called nodes) and edges. The properties of a graph describe various characteristics and attributes that can be associated with these components. The total number of vertices in a graph is known as its order. It represents the size of the graph. The total number of edges in a graph indicates the number of connections between the vertices. A network graph, also known as a graph or node-link diagram, represents a network or relationship between different elements. It consists of nodes (representing elements) connected by edges (representing relationships). By (Ali, et al, 2020). In computer science and mathematics, a graph is a basic data structure that is used to express relationships between items. It consists of a set of nodes, often referred to as vertices, joined together by edges. Graphs are effective model and problem-solving tools for a wide range of real-world issues, from computer networks and data structures to social networks and transportation systems (Smith, 2019).

In a graph, the nodes or vertices stand in for things or components, and the edges represent the links or connections that connect them. Depending on whether the edges have a specified direction or not, these connections might be either directional or undirected. An undirected edge denotes a two-way or symmetric interaction, while a directed edge suggests a one-way relationship. By (Ali, & Sajjad, 2019). The fundamental topic of graph theory is graphs. Since there is a clear connection between mathematics and chemical structure (what he called a chemical-graphical image), J. J. Sylvester first used the word "graph" in this sense in 1878. The structural features of benzenoid compounds and octahedra are combined to create benzenoid planar octahedron networks (Baig, et al, 2018). Similar to the structure of benzene, benzoids are hexagonally arranged, cyclic, planar hydrocarbon molecules. On the other hand, octahedra are polyhedra having eight faces, similar to a standard hexahedron.

When these two structures are combined, new planar networks are created (Trinajstic, 1992). The simplest benzenoid compound is benzene, which is made up of a hexagonal ring of carbon atoms with alternate single and double bonds. Because of the distinctive bonding structure that gives benzene its aromaticity, it is very stable and comparatively inert. Different substituents, such as methyl groups (CH3), hydroxyl groups (OH), or nitro groups

(NO2), among others, can be linked to the benzene ring(s) in benzenoid molecules. These substituents can alter the compound's reactivity, solubility, and biological activity, as well as its chemical and physical properties (Akhter, & Imran, 2016). The term "middle graph" describes the graph that represents the connectedness between the original network's edge midpoints. The middle graph in the instance of the Benzenoid Planar Octahedron Networks represents the connection between the midpoints of the Benzenoid octahedra's edges. The vertices of a middle graph can be compared to points on a line or in a one-dimensional space. The gaps between each pair of vertices are represented by the edges. Each edge represents an interval on the line and joins two vertices.

The intervals connected with the edges are non-overlapping and completely round the line. Chemical graph theory's investigation of topological aspects sheds light on the structural traits and connectivity patterns of molecular networks. The analysis of planar graphs in particular has drawn a lot of interest because of its importance in the field of organic chemistry. By (Aslam, et al, 2019). Among different types of planar graph structures, benzenoid planar octahedron networks have intriguing characteristics and have many uses in nanotechnology, drug design, and material science (Kulli, 2016). One or more benzene rings are found in the chemical compounds referred to as benzoids compounds. These substances function as the fundamental building blocks for a vast variety of organic molecules and are essential in several chemical reactions. Due to their stability and reactivity, benzene rings and their derivatives have garnered a lot of interest for their distinctive structure (Smith, 2019). The research of benzenoid planar octahedron networks has become more well-known recently. These networks provide a geometric representation of benzenoid molecules, allowing topological investigation into their characteristics. We can learn a lot about the aromaticity, electrical characteristics, and stability of benzenoid compounds by taking into account the connectivity and arrangement of the benzene rings in these networks. Middle graphs feature a number of intriguing characteristics. They can be described by their intersection model, which is one of their fundamental characteristics.

This indicates that an edge exists between both vertices in a middle graph if and only if the relevant intervals linked to the vertices cross. Due to this characteristic, middle graphs are advantageous for a number of uses, (Liu, et al, 2022). including computer science algorithms, interval graphs, and scheduling. A subfield of mathematics known as topology studies the characteristics of space that are maintained by continuous transformations. It concentrates on the characteristics of graphs that hold true when they are stretched, bent, or twisted when used in relation to graph theory. (Babar, et al, (2020). Analysis of connectedness, cycles, symmetry, planarity, and other structural traits that are invariant under specific transformations is necessary to understand the topological aspects of the Middle Graph of Benzenoid Planar Octahedron Networks. By (Arockiaraj, et al, 2016). The embedding of graphs (networks of nodes and edges) into surfaces is studied by topological graph theory, which also looks into the characteristics of these embeddings. It examines the connections between graph theory and topology and offers insights into topologically applicable subjects such as planar graphs, graph coloring, and graph algorithms. (Akhter, & Imran, 2016). While planar octahedron networks are made by combining normal octahedra in a planar configuration, benzenoids are organic compounds made of benzene rings connected by carbon-carbon bonds.

The middle graph represents a portion of the vertices and edges in these networks and offers important details about their structural properties. Materials science, chemistry, and nanotechnology are just a few industries that could benefit from understanding the topological characteristics of the Middle Graph of Benzenoid Planar Octahedron Networks. (Arockiaraj, et al, 2016). It offers perceptions of these particular networks' structural characteristics, which can help create and produce new materials with specific qualities or applications. To understand the complex structure and potential uses of the Middle Graph of Benzenoid Planar Octahedron Networks, researchers and mathematicians study these topological aspects using mathematical models, simulations on computers, and graph algorithms.

American mathematician Frank Harary is renowned for his significant contributions to graph theory. He researched on several different subjects, such as topological facets of graph theory. A classic in the discipline, Harary's book "Graph Theory" (1969) discusses many topological ideas associated with graphs. A Swiss mathematician named Euler made important advances in graph theory. He established the discipline of graph theory by solving the well-known Seven Bridges of Königsberg issue in 1736, which is regarded as one of the first findings in the field. (Liu, et al, 2019) Quantitative structure-property connection (QSPR) and quantitative structure-activity relationship (QSAR) research, in particular, heavily relies on molecular descriptors. An illustration of a molecular descriptor is a topological descriptor (Liu, et al, 2020). The usage of various topological indices in chemistry is one of the many contemporary uses. Based on the structural characteristics of the graphs used in their calculation, these variables can be classified. For instance, the Hosoya index is determined by counting a graph's non-incident edges (Azari, et al, 2013). The Randi connectivity index, (Randić, 2002). Zagreb group indices, and Estrada indices are also impacted by the graph's spectrum, degree of vertices, and vertex degree (Siddiqui, et al, 2016). The Wiener index, which is based on the topological distance of graph vertices, is probably the most popular and commonly used topological index. By (Mohar, et al, 1993). It was clarified, it was explained and put to use index topological. Harold Wiener used it to define in comparison the boiling points of various alkane isomers in 1947.

Chapter 2

LITERATURE REVIEW

This chapter introduced some basic concept and definition related to graph theory, Middle graph and Molecular graphs. They are most important for the present study to give a clear study.

2.1 Basic Definition of Theory

2.1.1 Graph

The two fundamental elements of a graph, a mathematical structure, are vertices (also known as and edges. A Graph G is referred to as an ordered pair (V, E). The collection of edges, which are the connection points or connections that join the vertices, is represented by the letter "E".

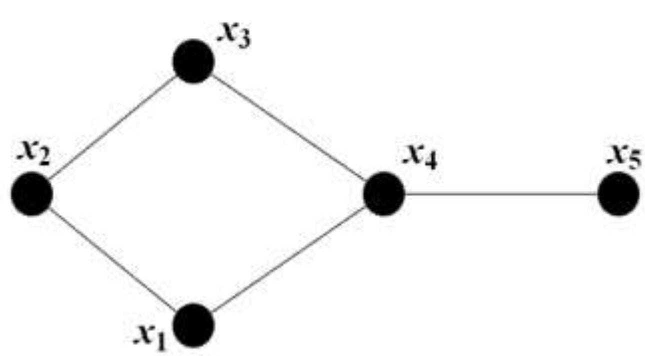


Figure: 2.1 Graph

2.1.2 Edge

In a graph, a link between two nodes or vertices is represented by an edge. It can be considered as a connection between two network elements.

Figure 2.2: Edge Graph

2.1.3 Vertex

A vertex, also is frequently referred to as a node, is the fundamental component of a graph. A graph is made up of a collection of vertices joined by edges.

Figure: 2.3 Vertex Graph

2.1.4 Line

Two point connects by a path is called line.

Figure: 2.4 Line Graph

2.1.5 POINT

A single dot on the graph represents a single symbol of each data point in a point graph.

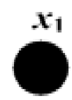


Figure: 2.5 Point Graph

2.1.6 Adjacent Vertices

Adjacent vertices represent two vertices that are directly related to one another through an edge in the concept of graph theory.

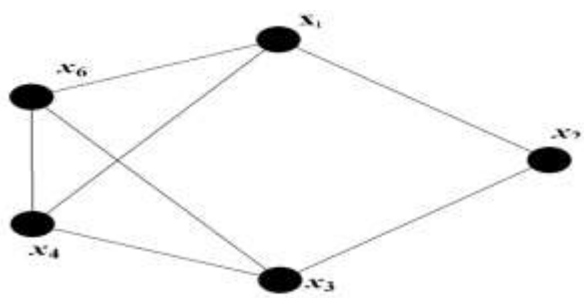


Figure: 2.6 Adjacent Graph

2.1.7 Isolated Vertex

A vertex that is not connected to any other vertex in a graph is known as an isolated vertex in the context of graph theory.

Figure: 2.7 Isolated Graph

In this case, the vertices "a'' and "b" are not connected to one another or to any other vertices. Therefore, vertices "a" and "b" have zero degree.

2.1.8 Order of a Graph

The quantity of vertices or nodes in a graph is referred to as the graph's order. It provides information on the size or total number of graph elements

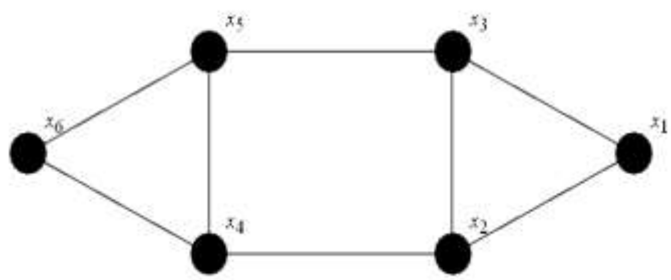


Figure: 2.8 Order of Graph

V= $\{x_{1,}x_2, x_{3,}x_{4,}x_{5,}x_6\}$

V=6

2.1.9 Graph Size

The number of edges in a graph determines its size.

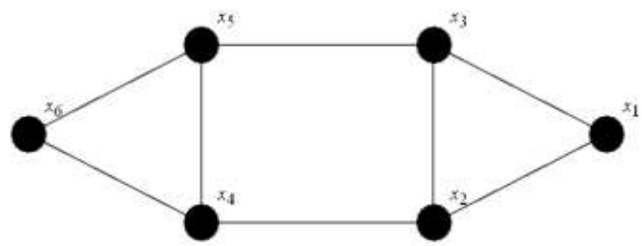


Figure: 2.9 Size Graph

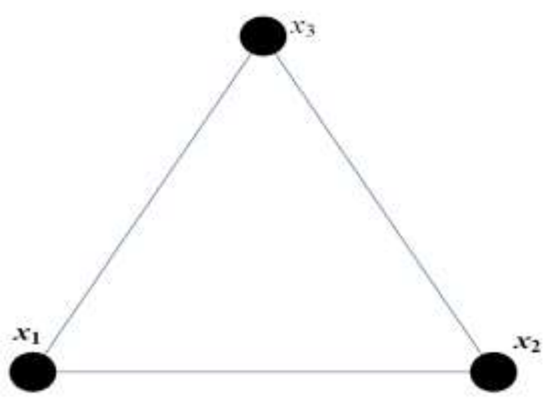
E= $\{(x_{1,}x_2),(x_{1,}x_3),(x_{3,}x_2),(x_{3,}x_5),(x_{2,}x_4),(x_{5,}x_4),(x_{5,}x_6),(x_{4,}x_6)\}$

E=8

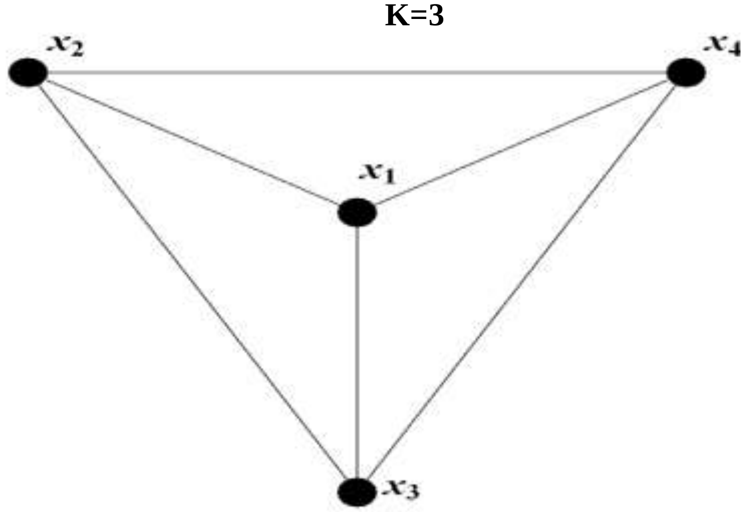
2.1.10 Complete Graph

A full network has no isolated nodes or disconnected segments. It is typically depicted as a group of vertices, or dots, joined by edges, or lines.

K=2



K=3



k=4

Figure: 2.10 Complete graph

2.1.11 Regular Graph

In a regular graph, also known as a degree graph, there are the same number of neighbors at each vertex. To put it another way, all vertices in a regular network have the same degree.

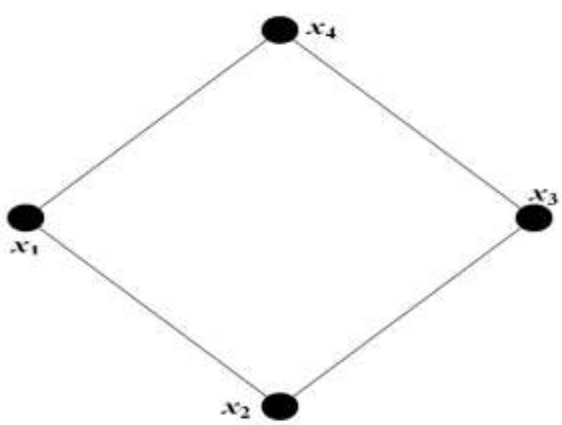


Figure: 2.11 Regular graph

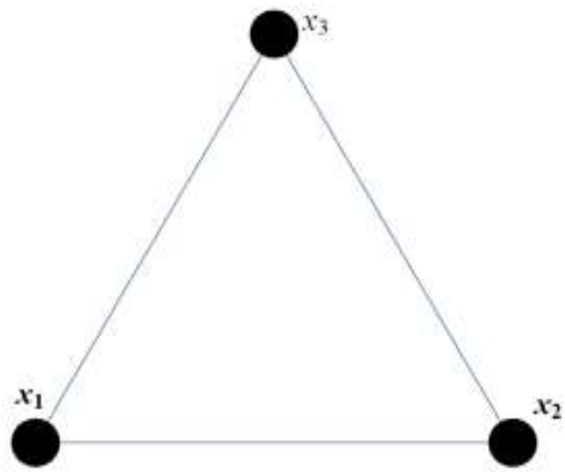


Figure: 2.11 Regular graph

2.1.12 Null Graph

A graph with the formula Each vertex has the same number of neighbors in the equation

G = (V, E), where V denotes the set of vertices and E denotes the set of edges. referred to as a null graph. Since V and E are both empty sets in this case, a null graph has no elements.

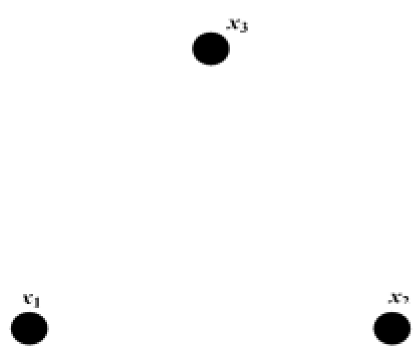


Figure: 2.12 Null Graph

2.1.13 Directed Graph

In a directed graph, the edges are typically represented as arrows pointing from one node to the next. The relationship between the nodes and their corresponding directions are depicted by these arrows. If there is an edge between the two nodes, for example, a directed connection exists from node A to node B, but not necessarily from B to A.

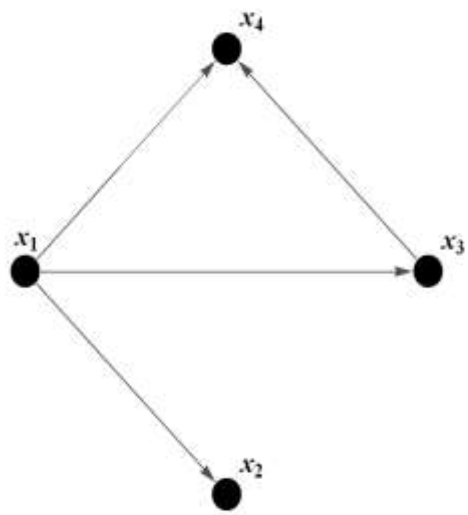


Figure: 2.13 Directed Graph

See above the graph has four vertices $\{X_1, X_2, X_3, X_4\}$ and four edges $\{(X_1, X_2)$, (X_1, X_3), (X_1, X_4), (X_3, X_4). So, it is a directed graph, each edge has marked with an arrow indicating its direction. In directed graph, (a, b) is distinct from (b, a).

2.1.14 Undirected Graph

In a graph with no direction a pair (V, E) can be used to define G, where V stands for a group of vertices and E for a group of edges. The edges of an undirected graph are pairs of unordered vertices. If (u, v) is an edge in the graph E, then there is a connection between the nodes u and v.

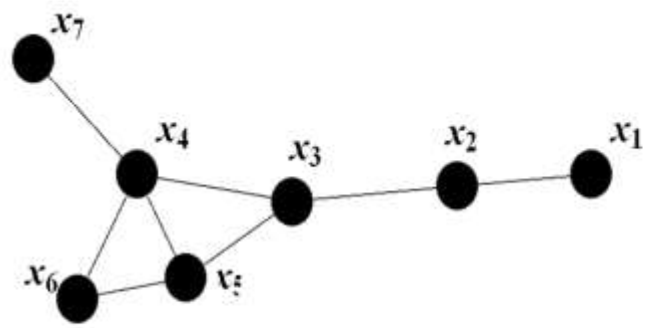


Figure: 2.14 Undirected Graph

2.1.15 Connected Graph

Every pair of vertices in a connected network has a path that connects them. In a connected network, any two vertices have a set of edges that can be used to move from one to the other.

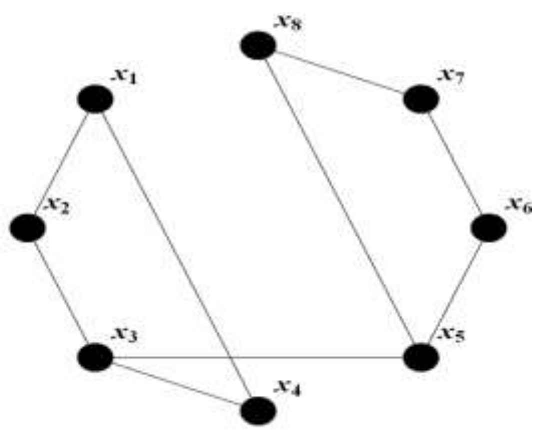


Figure: 2.15 Connected Graph

2.1.16 Disconnected Graph

When there are multiple pieces or subgraphs that are not connected to one another, the graph is said to be disconnected or disjoint

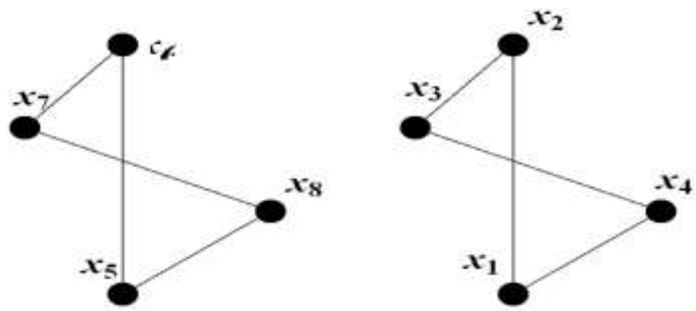


Figure: 2.16 Disconnected Graph

2.1.17 Simple Graph

There is only one edge that connects any two vertices in a simple graph and each edge connects two distinct vertices. This means that neither a self-loop which connects a vertex to itself nor many edges which connect the same set of vertices exist.

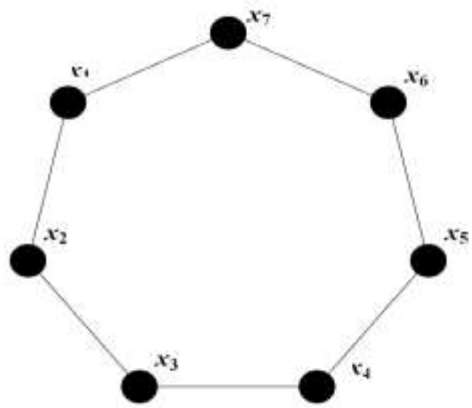


Figure: 2.17 Simple Graph

2.1.18 Middle Graph

The graph whose is V$|G|$E$|G|$ is the middle graph M[G] of a graph. Two vertices are adjacent in this graph only if and only if they are either adjacent edges of G or one is a vertex and the other is an edge incident on it. G is denoted by M$|G|$

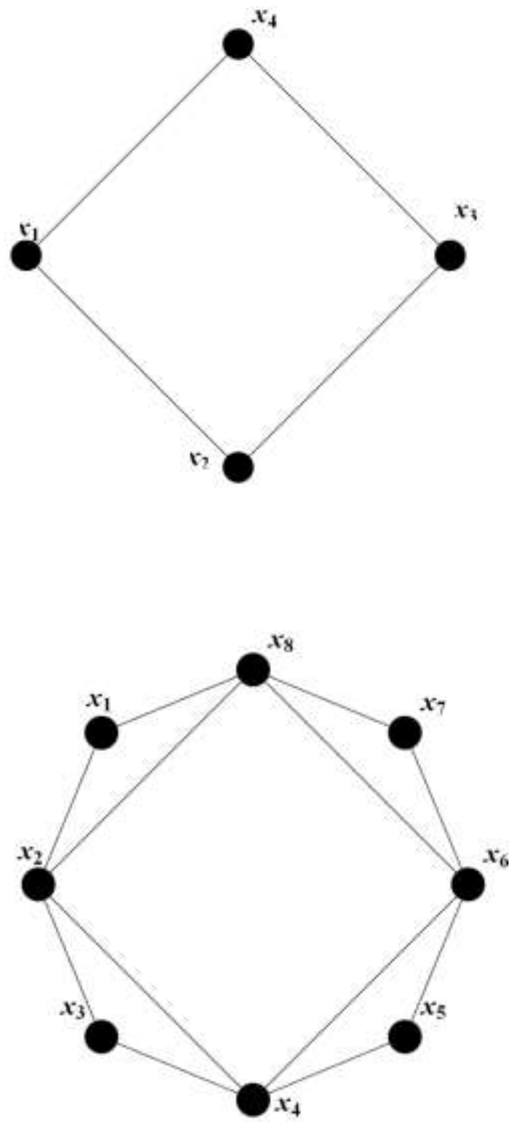


Figure: 2.18 Middle Graph

2.1.19 LOOP

A single edge that starts and ends at the same vertex serves as the representation for a loop. It simply connects a vertex back to itself, creating a "loop" within the graph.

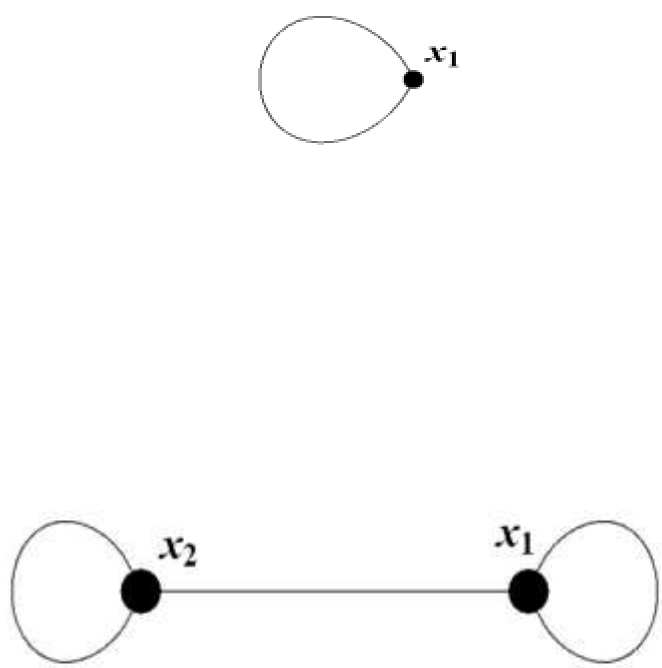


Figure: 2.19 Loop

2.1.20 Cycle

Except for the start and last vertices, which are attached to each other in order to complete the cycle, every vertex in a cycle graph has connections to exactly two other vertices.

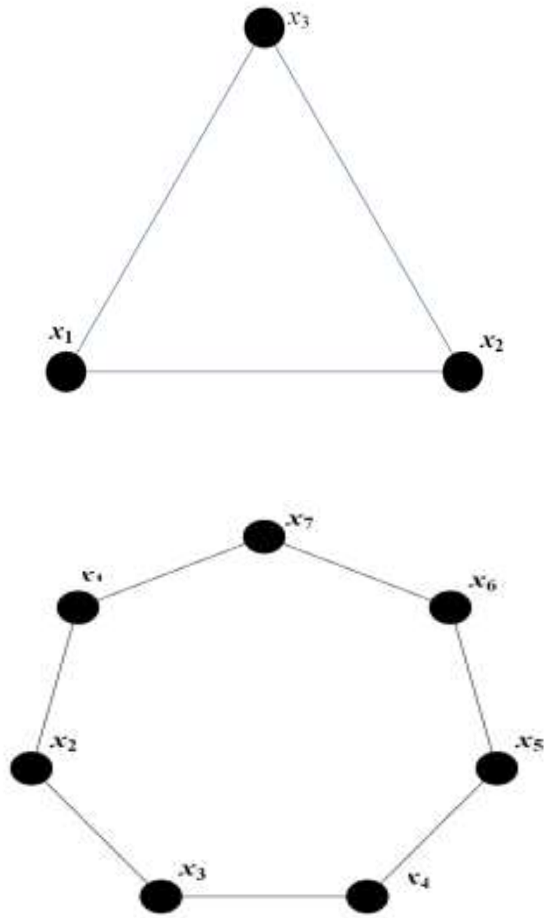


Figure: 2.20 Cycle Graph

2.1.21 Cyclic Graph

A graph is said to be cyclic if it has at least one cycle, or a path that starts and finishes at the same node.

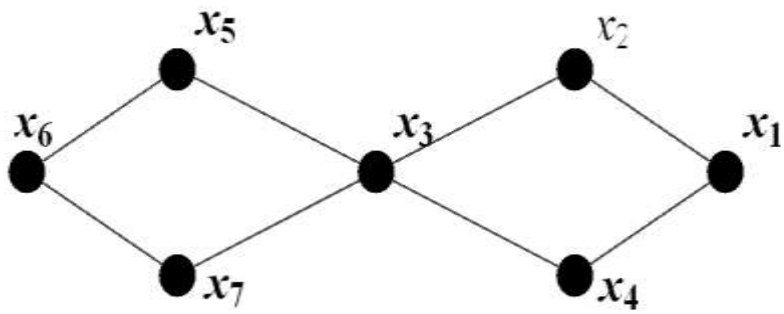


Figure: 2.21 Cyclic Graph

2.1.22 Acyclic Graph

A graph is said to be acyclic if there are no graph cycles. There are simply two sides to an acyclic graph.

A line graph is constructed by connecting a number of points with lines to create a graphical version of data.

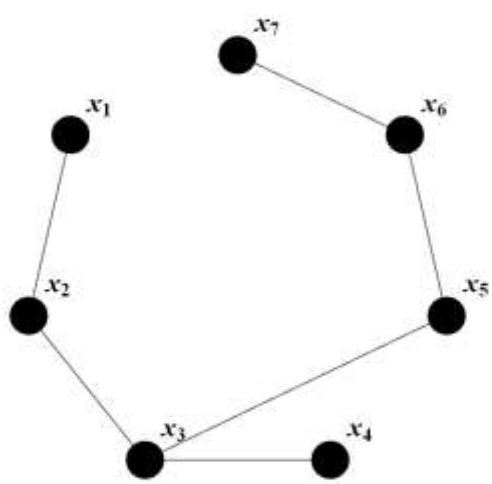


Figure: 2.22 acyclic graph

2.1.23 Degree of a vertex

The total number of edges connecting a vertex is specified as the vertex degree in graph theory. The degree vertex will be two if it contains a loop.

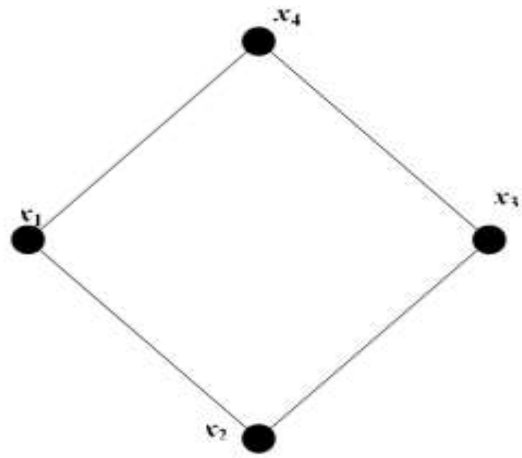


Figure: 2.23 Degree of vertex Graph

2.1.24 Sum of Degree of Vertices

The number of edges that are connected to a vertex determines its degree. According to the degree sum formula, the total degree of all the vertices in a (finite) graph equals a twice as many edges as there are in the graph.

It is denoted by S_m and defined as

$$\sum_{n \in nH(m)} dn$$

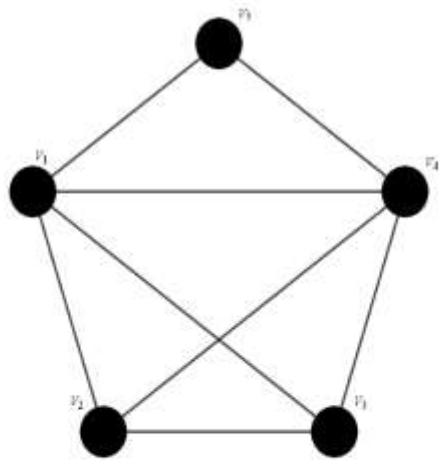
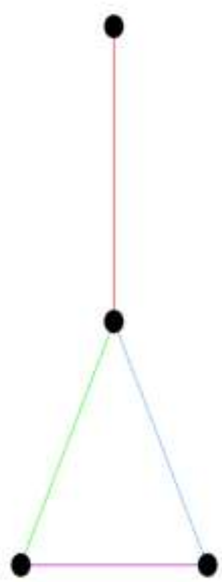
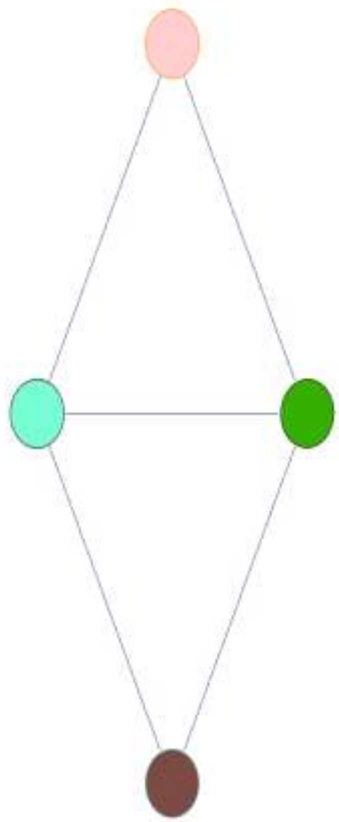


Figure: 2.24 Some of degree vertex Graph

2.1.25 Line Graph

Vertex set V(G) and edge set E (G) are present in any graph G. and define with vertex set L(G)=E(G).





L(G)

Figure: 2.25 Line graph

2.1.26 Trivial Graph

A graph with one vertex is referred to as a trivial graph. Otherwise graph is called non trivial graph.

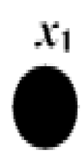


Figure: 2.26 Trivial Graph

2.1.27 Molecular Graph

A molecular graph is a specific type of graph that shows the structure of a molecule. These bonds and their interactions are shown in a visual and structural manner by the molecular graph. In chemistry, atoms come together to form molecules. In a molecular graph, atoms are shown as vertices, while the relationships between them are shown as edges. The edges of the graph, which represent the atoms in the molecule's vertices, show the chemical bonds forming between them.

2.1.28 Open walk

A walk with a distinct initial vertex and terminal vertex is known as an "open walk" the open walk is $(x_1,\ x_{2,}\ x_{3,}\ x_4, x_5, x_6)$.

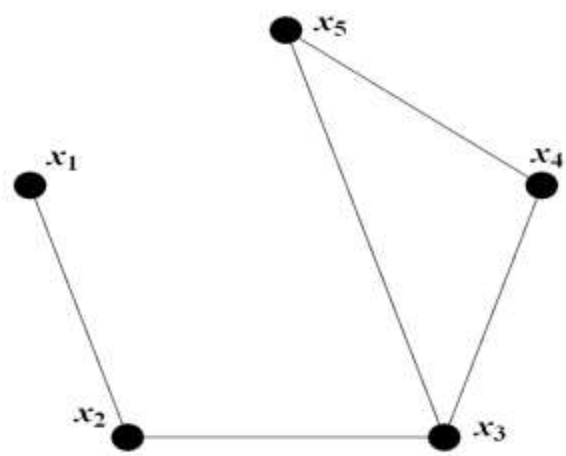


Figure: 2.28 Open walk

2.1.29 Closed walk

A closed walk with same initial and terminal vertex is called closed walk.

$(x_1,\ x_{2,}\ x_{3,}\ x_4, x_5, x_6\ )$ is closed walk.

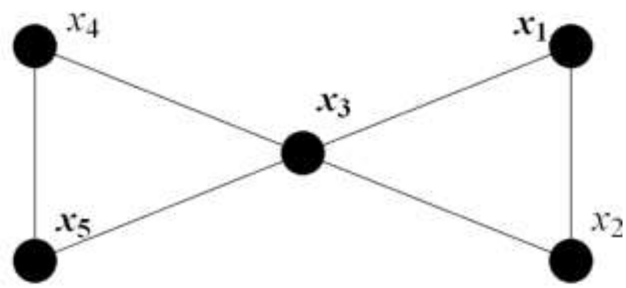


Figure: 2.29 closed walk

2.1.30 Distance of Graph

The number of edges connecting any two vertices in a graph that may be reached via the shortest path is called the graph distance; in the example above, that distance is between x_1 and x_4 .

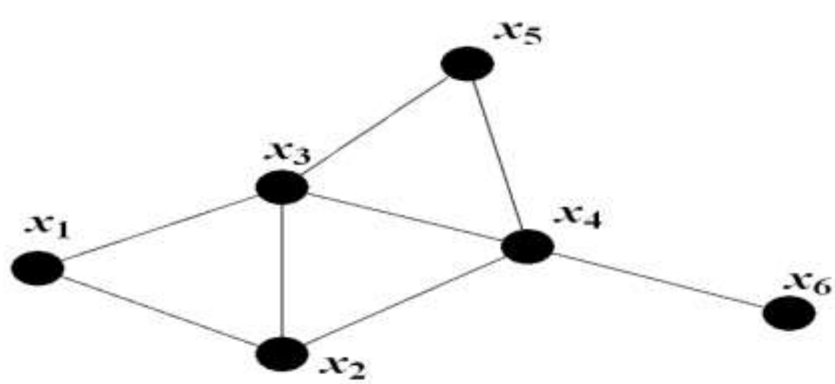


Figure: 2.30 Distance of Graph

2.1.31 Diameter of a graph

We will look at all of the paths in the graph, select the largest one, and count the number of edges that connect it to any two vertices. Diameter is defined as the total number of edges between any two vertices connected by the largest path.

In the above example, the maximum path is $x_1, x_{2,} x_{4,} x_6$ so its diameter is 3.

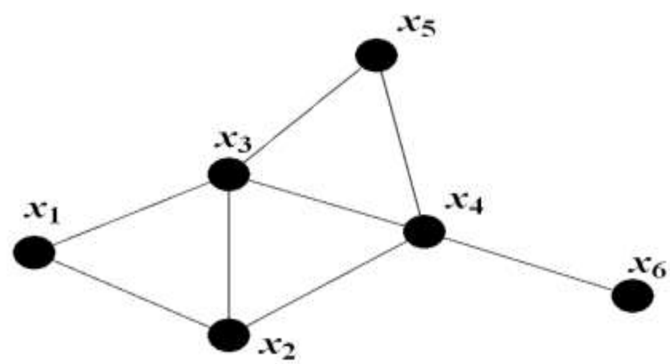


Figure: 2.31 Diameter of Graph

2.1.32 Girth of a Graph

The shortest closed walk around a graph with the same start and finish vertices but no repeated edges or vertices is said to have the girth of the graph. It is the total number of edges in a graph minimal cycle, and put it another way. The shortest cycle present in the above graph is $x_1, x_{2,} x_{3,}$ so its girth is 2.

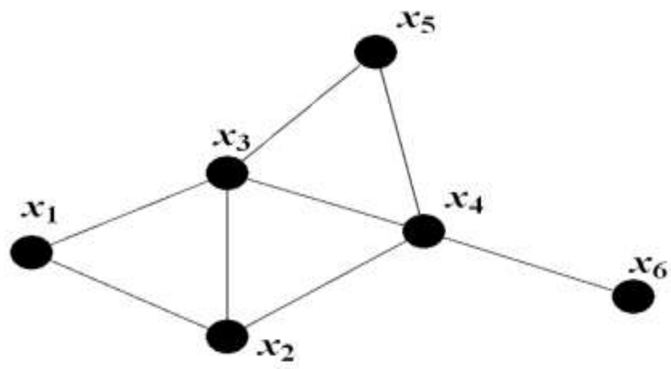


Figure: 2.32 Girth of Graph

2.1.33 Construction of Line Graph

A line graph is constructed by connecting a number of points with lines to create a graphical version of data.

2.1.34 Applications of Graph Theory

Many different fields have utilized graph theory in a variety of ways. Here are a few important examples:

- Traffic Networks
- Computer Science
- Physics and Chemistry
- Electrical Engineering
- Social Science
- General
- Operations Research and Optimization
- Networking and Telecommunications
- Computational and Bioinformatics Biology
- Analysis of Social Networks
- Logistics and Transportation
- Image and Pattern Recognition

- VLSI Design and Chip Layout
- Algorithm Design and Analysis
- Internet and Web Applications
- Network security and cryptography;
- Data mining and machine learning
- Autonomous systems and robotics
- Networks of Electrical Power
- Supply Chain Administration

Chapter 3

RESEARCH METHODOLOGY

3.1 Introduction

The study of graph theory is used to calculate topological indices, and these are useful tools. A type of graph theory called the chemical graph has numerous uses in both mathematics and science (Imran, et al, 2016). A good illustration of this is a molecular graph. The structural study of molecular graphs, networks, and Topological indices are quantifiable values that are connected to graphs or chemical structures and reveal details about their topological characteristics. These indices are frequently used to characterize and forecast the characteristics and behavior of molecules or networks in a variety of disciplines, including chemistry, physics, and computer science. Topological indices are used in chemistry to measure the links between a compound's structure and properties. They come from how molecules are represented as graphs, where atoms are represented as vertices and bonds between atoms as edges.

Topological indices can shed light on a variety of qualities, including boiling point, solubility, biological activity, and more, by examining the connection and arrangement of the atoms in the molecule (Smith, 2019). There are numerous widely used topological indices, each of which captures various aspects. Topological indices that depend on distance, degree, and polynomials are some of the most used varieties. Molecules and molecular compounds are frequently described using chemical graphs. trees have lately been shown to depend heavily on topological indices, as many researchers have found. There are multiple topological indices that are often employed, and they each capture different aspects of molecule structure. Examples include, among others, the Balaban index, the Wiener index, the Randi index, and the Zagreb indices (Mohar, et al, 1993). These indices are computed using mathematical methods that take into account a variety of graph-theoretical aspects, including vertex degrees, distances between vertices, and graph symmetries.

The following topological indices were used.

- Randić Index $R(G_1)$
- Harmonic index H (G_1)
- First Zagreb index $N_1(G_1)$

- 2nd Zagreb index $N_2(G_1)$
- Hyper Zagreb index HN (G_1)
- Sum connectivity index SCI (G_1)
- Forgotten index F (G_1)
- 1st Reclassified Zagreb index RE$G_1(G_1)$
- 2nd Reclassified Zagreb index RE$G_2(G_1)$
- 3rd Reclassified Zagreb index RE$G_3(G_1)$
- Geometric-Arithmetic index G$A_5(G_1)$
- Inverse sum index ISI (G_1)
- AB$C_4(G_1)$ $index$

3.1.1 Randić Index $\boldsymbol{R}(\boldsymbol{G_1})$

The Randi'c index, developed by Milan Randi'c,in 1975 is the most popular, and thoroughly studied degree-based topological indicator. Randic's index, sometimes referred to as the product connectivity index, is one of the best molecular identifiers used in QSPR and QSAR research and is suitable for determining the degree of the branch of the carbon-atom skeleton of saturated hydrocarbons. It belongs to the first generation of degree-based topological indices. By (Randić, 2002).

$$R(G_1)= \sum_{u_{i,}v_j \in E_{G_1}} \frac{1}{\sqrt{\rho_{u_{i,}}\rho_{v_j}}}$$

3.1.2 Harmonic index $\boldsymbol{H}(\boldsymbol{G_1})$

when this Harmonic index originally appeared. The relationship between the harmonic index and the graph eigenvalues were taken into consideration by Favaron et al. Here, we will discuss the related extremal graphs and study the minimum and maximum harmonic index values for simple connected graphs and trees. (Ranjini, et al, 2013). The harmonic index, G, of a graph, is denoted by

$$H(G_1)=\sum_{u_{i,}v_j\in E_{G_1}}\frac{2}{\rho_{u_{i,}}+\rho_{v_j}}$$

3.1.3 First Zagreb index and 2nd Zagreb index:

The graph theory concept known as the Zagreb Index is used to quantify the complexity or molecular branching of organic molecules (Liu, et al, 2019). It is called after the Croatian capital city of Zagreb and was first proposed by Croatian mathematicians. Gutman and Trinajstic proposed the first Zagreb index in 1972, and it includes calculating the carbon atom skeleton's branching. First and secondlyThe first Zagreb formula and the Zagreb indices are as follows.

$$N_1(G_1)=\sum_{u_{i,}v_j\in E_{G_1}}[\rho_{u_{i,}}+\rho_{v_j}]$$

$$N_2(G_1)=\sum_{u_{i,}v_j\in E_{G_1}}[\rho_{u_{i,}}\times\rho_{v_j}]$$

3.1.4 Hyper Zagreb index $HN(G_1)$

In the family of Zagreb indices, the hyper-Zagreb index is a significant branch that has been the subject of research. The extremal graphs between vertex trees (acyclic), unicyclic, and bicyclic graphs are found for the hyper Zagreb index using these lovely mathematical features (Wang, & Wei, 2015). Additionally, these graphs' hyper-Zagreb indices' sharp upper and lower boundaries are given.

$$H\,\mathrm{N}\,(G_1)=\sum_{u_{i,}v_j\in E_{G_1}}[\rho_{u_{i,}}+\rho_{v_j}]^2$$

3.1.5 Sum connectivity index SCI(K)

A molecule's complexity and branching are measured by the sum connectivity index. It has been discovered to correlate with a number of molecular physicochemical and biological characteristics, making it valuable for predicting molecular activities and attributes in drug design, toxicity prediction, and other medicinal chemistry applications (Falahati, & Azari, 2016). Zhou and Trinajstic were the ones who first proposed sum-connectivity. The following definition applies to the graph G's sum-connectivity index:

$$SCI(G_1)= \sum_{u_{i,} v_j \in E_{G_1}} \frac{1}{\sqrt{\rho_{u_{i,}} + \rho_{v_j}}}$$

3.1.6 Forgotten index

The sum of the cubes of the vertex degree vertices in a network is known as the Forgotten topological index, or F-index. We only take into account the intermolecular forces between the atoms of a molecule when calculating the classical F-index, ignoring the intermolecular forces that occur between the atoms and bonds. (Furtula, & Gutman, 2015). The forgotten index, often known as the F-index, was found by Furtula and Ivan Gutman and is defined as

$$F(G_1)= \sum_{u_{i,} v_j \in E_{G_1}} [\, (\rho_{u_{i,}})^2 (\rho_{v_j})^2]$$

3.1.7 1st ,2nd and 3rd Reclassified Zagreb index

Reclassified Zagreb indexes come in a variety of forms, each with a unique reclassification methodology. The reclassified first Zagreb index (R1), reclassified second Zagreb index (R2), and reclassified third Zagreb index (R3) are a few examples of commonly used reclassified Zagreb indices (Wang, & Wei, 2015). These indices compute a modified Zagreb index by accounting for the types of atoms or bonds, atom degrees, and other structural characteristics. Ranjini et al. introduced the Redefined Zagreb index, which is defined as

$$ReN_1(G_1)= \sum_{u_{i,} v_j \in E_{G_1}} \frac{\rho_{u_{i,}} \times \rho_{v_j}}{\rho_{u_{i,}} + \rho_{v_j}}$$

$$ReN_2(G_1)= \sum_{u_{i,} v_j \in E_{G_1}} \frac{\rho_{u_{i,}} + \rho_{v_j}}{\rho_{u_{i,}} \times \rho_{v_j}}$$

$$ReN_3(G_1)= \sum_{u_{i,} v_j \in E_{G_1}} (\rho_{u_{i,}} \times \rho_{v_j})(\, \rho_{u_{i,}} + \rho_{v_j})$$

3.1.8 Geometric Arithmetic - index

The GA index has uses in chemoinformatic, drug design, and quantitative structure-activity relationship (QSAR) investigations (Graovac, et al, 2011). Researchers can compare and analyses the structural characteristics of various compounds by calculating the GA index, and perhaps make predictions about their traits or activities. By Vukicevic and Furtula, the Geometric-Arithmetic index has been established.

$$GA_5(G_1) = \sum_{u_i, v_j \in E_{G_1}} \frac{2\sqrt{S_{u_i,} S_{v_j}}}{S_{u_i,} + S_{v_j}}$$

3.1.9 Inverse sum index

A topological statistic called the "Inverse Sum Index" is used to describe the structural and connectivity characteristics of graphs or molecular structures. The inverse sum of the distances between each pair of vertices in the graph is used to calculate it. Its symbol is ISI. By (Falahati, & Azari, 2016).

$$ISI(G_1) = \sum_{u_i, v_j \in E_{G_1}} \frac{\rho_{u_i,} \times \rho_{v_j}}{\rho_{u_i,} + \rho_{v_j}}$$

3.1.10 General inverse sum index

The general inverse sum index, denoted by ISI (a, b) (widet I ldeG), was recently proposed by Buragohain et al. and is defined as:

$$ISI_{(a,b)}(G_1) = \sum_{u_i v_j \in E(G)} \left[d_{ui} \times d_{vj} \right]^a \left[d_{ui} \times d_{vj} \right]^b$$

Where a and b are some real numbers.

3.1.11 Balaban index

Chemistry uses the Balaban index, a mathematical description, to rate the topological complexity of a molecule. It was first presented in 1989 by mathematicians Aleksandar Balaban and Ivan Gutman, who is also a chemist. In honor of Balaban's contributions to the field of chemical graph theory, the index bears his name.

The formula is for a graph G with 'n' vertices and ’m' edges.

$$B(G_1) = \left(\frac{m}{m-n+2}\right)\sum_{u_{i,}v_j \in E_{G_1}} \frac{1}{\sqrt{\rho_{u_{i,}} \times \rho_{v_j}}}$$

3.1.10 AB$C_4(G_1)$

If we are able to figure out how these connectivity chemical networks edges are divided up, which is done by adding up the degrees at which each edge's terminating vertices in each of these graphs are, then we can figure out how they are divided up., we can compute ABC4 indices (Falahati, & Azari, 2016). The ABC4 index was developed by Ghorbani et al. and is defined as

$$GA_5(G_1) = \sum_{u_{i,}v_j \in E_{G_1}} \frac{\sqrt{S_{u_{i+}}\, S_{v_{j-}}\, {}^2}}{S_{u_{i,}} . S_{v_j}}$$

Chapter 4

RESULTS AND DISCUSSIONS

SECTION 1

Theorem 4.1: Let the middle graph of benzenoid planar Octahedron networks MBPOH; forgotten index is equal to

$$F(MBPOH) = 118268$$

Proof: Let G_1 be the middle graph benzenoid planar octahedron network MBPOH, use Fig.1 , and the edges set of G_1 depending on the degree of the end vertices is divided into thirteen parts.

Table:1 Edge partition of degree based for MBPOH.

$\rho(u_i), \rho(v_j)$	Number of edges	$\rho(u_i), \rho(v_j)$	Number of edges
(3,4)	**144**	**(6,10)**	**192**
(3,6)	**288**	**(8,9)**	**24**
(4,6)	**168**	**(8,10)**	**96**
(4,9)	**24**	**(9,9)**	**12**
(6,6)	**216**	**(9,10)**	**24**
(6,8)	**120**	**(10,10)**	**36**
(6,9)	**96**		

Using Table 1 edge partition and the forgotten index, we can get the following result.

$$F(G_1) = 25|E_1(G_1)| + 45|E_2(G_1)| + 52|E_3(G_1)| + 97|E_4(G_1)| + 72|E_5(G_1)| + 100|E_6(G_1)| + 117|E_7(G_1)| \\ +136|E_8(G_1)| + 145|E_9(G_1)| + 164|E_{10}(G_1)| + 162|E_{11}(G_1)| + 181|E_{12}(G_1)| + 200|E_{13}(G_1)|$$

$$F(G_1) = 25(144) + 45(288) + 52(168) + 97(24) + 72(216) + 100(120) + 117(96) \\ +136(192) + 145(24) + 164(96) + 162(12) + 181(24) + 200(36)$$

We find the result after calculation

$$\Rightarrow F(MBPOH) = 118268$$

Theorem:4.2 Let the middle graph benzenoid planar Octahedron networks G_1,then we have

$$\operatorname{Re} ZG_1(G_1) = \frac{1070136}{455} + \frac{420218}{255} + \frac{5530}{19}$$

$$\operatorname{Re} ZG_2(G_1) = \frac{4737}{5}$$

$$\operatorname{Re} ZG_3(G_1) = 844488$$

Proof: Consider G_1 representing the middle graph of a benzenoid planar octahedron network (MBPOH). Using the edge partition and the reclassified Zagreb indices, we can obtain the results from Table 1.

$$\begin{aligned}\operatorname{Re} ZG_1(G_1) = {} & \frac{12}{7}|E_1(G_1)| + \frac{18}{9}|E_2(G_1)| + \frac{24}{10}|E_3(G_1)| + \frac{36}{13}|E_4(G_1)| + \frac{36}{12}|E_5(G_1)| + \frac{48}{14}|E_6(G_1)| \\ & + \frac{54}{15}|E_7(G_1)| + \frac{60}{16}|E_8(G_1)| + \frac{72}{17}|E_9(G_1)| + \frac{80}{18}|E_{10}(G_1)| \\ & + \frac{81}{18}|E_{11}(G_1)| + \frac{90}{19}|E_{12}(G_1)| + \frac{100}{20}|E_{13}(G_1)|\end{aligned}$$

$$\begin{aligned}\operatorname{Re} ZG_1(G_1) = {} & \frac{12}{7}(144) + \frac{18}{9}(288) + \frac{24}{10}(168) + \frac{36}{13}(24) + \frac{36}{12}(216) + \frac{48}{14}(120) + \frac{54}{15}(96) \\ & + \frac{60}{16}(192) + \frac{72}{17}(24) + \frac{80}{18}(96) + \frac{81}{18}(12) + \frac{90}{19}(24) + \frac{100}{20}(36)\end{aligned}$$

We find the result after calculation

$$\Rightarrow \operatorname{Re} ZG_1(G_1) = \frac{1070136}{455} + \frac{420218}{255} + \frac{5530}{19}$$

The ReZG_2 can be calculated by using 2nd the reclassified Zagreb indices as follows the result of table 1.

$$\operatorname{Re} ZG_2(G_1) = \frac{4737}{5}$$

$$\operatorname{Re} ZG_2(G_1) = \frac{7}{12}|E_1(G_1)| + \frac{9}{18}|E_2(G_1)| + \frac{10}{24}|E_3(G_1)| + \frac{13}{36}|E_4(G_1)| + \frac{12}{36}|E_5(G_1)| + \frac{14}{48}|E_6(G_1)|$$
$$+ \frac{15}{54}|E_7(G_1)| + \frac{16}{60}|E_8(G_1)| + \frac{17}{72}|E_9(G_1)| + \frac{18}{80}|E_{10}(G_1)| + \frac{18}{81}|E_{11}(G_1)| + \frac{19}{90}|E_{12}(G_1)|$$
$$+ \frac{20}{100}|E_{13}(G_1)|$$

$$\operatorname{Re} ZG_2(G_1) = \frac{7}{12}(144) + \frac{9}{18}(288) + \frac{10}{24}(168) + \frac{13}{36}(24) + \frac{12}{36}(216) + \frac{14}{48}(120) + \frac{15}{54}(96)$$
$$+ \frac{16}{60}(192) + \frac{17}{72}(24) + \frac{18}{80}(96) + \frac{18}{81}(12) + \frac{19}{90}(24) + \frac{20}{100}(36)$$

We find the result after calculation

$$\Rightarrow \operatorname{Re} ZG_2(G_1) = \frac{4737}{5}$$

The ReZG_3 can be calculated by using 3rd the reclassified Zagreb indices as follows the result of table 1.

$$\operatorname{Re} ZG_3(G_1) = 844488$$

$$\operatorname{Re} ZG_3(G_1) = 84|E_1(G_1)| + 162|E_2(G_1)| + 240|E_3(G_1)| + 468|E_4(G_1)| + 432|E_5(G_1)|$$
$$+672|E_6(G_1)| + 810|E_7(G_1)| + 960|E_8(G_1)| + 1224|E_9(G_1)| + 1440|E_{10}(G_1)|$$
$$+1458|E_{11}(G_1)| + 1710|E_{12}(G_1)| + 2000|E_{13}(G_1)|$$

$$\operatorname{Re} ZG_3(G_1) = 84(144) + 162(288) + 240(168) + 468(24) + 432(216)$$
$$+672(120) + 810(96) + 960(192) + 1224(24) + 1440(96)$$
$$+1458(12) + 1710(24) + 2000(36)$$

We find the result after calculation

$$\Rightarrow \operatorname{Re} ZG_3(G_1) = 844488$$

Theorem 4.3: Let G_1 be the middle graph benzenoid planar octahedron network MBPOH; its Randic index is

$$\mathrm{R}(G_1)=\frac{1}{\sqrt{12}}(144)+\frac{1}{\sqrt{18}}(288)+\frac{1}{\sqrt{24}}(168)+\frac{1}{\sqrt{36}}(24)+\frac{1}{\sqrt{36}}(216)$$
$$+\frac{1}{\sqrt{48}}(120)+\frac{1}{\sqrt{54}}(96)+\frac{1}{\sqrt{60}}(192)+\frac{1}{\sqrt{72}}(24)+\frac{1}{\sqrt{80}}(196)$$
$$+\frac{1}{\sqrt{81}}(12)+\frac{1}{\sqrt{90}}(24)+\frac{1}{\sqrt{100}}(36)$$

Proof: Let can be calculated by using Randic index

$$\mathrm{R}(G_1)=\frac{1}{\sqrt{12}}|E_1(G_1)|+\frac{1}{\sqrt{18}}|E_2(G_1)|+\frac{1}{\sqrt{24}}|E_3(G_1)|+\frac{1}{\sqrt{36}}|E_4(G_1)|+\frac{1}{\sqrt{36}}|E_5(G_1)|$$
$$+\frac{1}{\sqrt{48}}|E_6(G_1)|+\frac{1}{\sqrt{54}}|E_7(G_1)|+\frac{1}{\sqrt{60}}|E_8(G_1)|+\frac{1}{\sqrt{72}}|E_9(G_1)|+\frac{1}{\sqrt{80}}|E_{10}(G_1)|$$
$$+\frac{1}{\sqrt{81}}|E_{11}(G_1)|+\frac{1}{\sqrt{90}}|E_{12}(G_1)|+\frac{1}{\sqrt{100}}|E_{13}(G_1)|$$

We find the result after calculation by using table 1.

$$\mathrm{R}(G_1)=\frac{1}{\sqrt{12}}(144)+\frac{1}{\sqrt{18}}(288)+\frac{1}{\sqrt{24}}(168)+\frac{1}{\sqrt{36}}(24)+\frac{1}{\sqrt{36}}(216)$$
$$+\frac{1}{\sqrt{48}}(120)+\frac{1}{\sqrt{54}}(96)+\frac{1}{\sqrt{60}}(192)+\frac{1}{\sqrt{72}}(24)+\frac{1}{\sqrt{80}}(196)$$
$$+\frac{1}{\sqrt{81}}(12)+\frac{1}{\sqrt{90}}(24)+\frac{1}{\sqrt{100}}(36)$$

□

Theorem 4.4: Let G_1 *be* the middle graph benzenoid planar octahedron network MBPOH; its ABC_4 index GA_5

$$\bullet ABC_4(G_1)=\frac{\sqrt{15}}{6}(144)+\frac{\sqrt{14}}{6}(288)+\frac{\sqrt{3}}{3}(168)+\frac{\sqrt{11}}{6}(24)+\frac{\sqrt{10}}{6}(216)$$
$$+\frac{1}{2}(120)+\frac{\sqrt{78}}{18}(96)+\frac{\sqrt{210}}{30}(192)+\frac{\sqrt{30}}{12}(24)+\frac{\sqrt{5}}{5}(96)$$
$$+\frac{4}{9}(12)+\frac{\sqrt{170}}{30}(24)+\frac{2\sqrt{36}}{10}(36)$$

$$\bullet GA_5(G_1)=\frac{4\sqrt{3}}{7}(144)+\frac{2\sqrt{2}}{3}(288)+\frac{2\sqrt{6}}{5}(168)+\frac{12}{13}(24)+1(216)$$
$$+\frac{4\sqrt{3}}{7}(120)+\frac{2\sqrt{6}}{5}(96)+\frac{\sqrt{15}}{4}(192)+\frac{12\sqrt{2}}{17}(24)+\frac{4\sqrt{5}}{9}(96)$$
$$+1(12)+\frac{6\sqrt{10}}{19}(24)+1(36)$$

Proof: The ABC_4(G_1) can be calculated by using atom-bond connectivity as follows

$$ABC_4(G_1)=\frac{\sqrt{15}}{6}\left|E_1(G_1)\right|+\frac{\sqrt{14}}{6}\left|E_2(G_1)\right|+\frac{\sqrt{3}}{3}\left|E_3(G_1)\right|+\frac{\sqrt{11}}{6}\left|E_4(G_1)\right|+\frac{\sqrt{10}}{6}\left|E_5(G_1)\right|$$
$$+\frac{1}{2}\left|E_6(G_1)\right|+\frac{\sqrt{78}}{18}\left|E_7(G_1)\right|+\frac{\sqrt{210}}{30}\left|E_8(G_1)\right|+\frac{\sqrt{30}}{12}\left|E_9(G_1)\right|+\frac{\sqrt{5}}{5}\left|E_{10}(G_1)\right|$$
$$+\frac{4}{9}\left|E_{11}(G_1)\right|+\frac{\sqrt{170}}{30}\left|E_{12}(G_1)\right|+\frac{3\sqrt{2}}{10}\left|E_{13}(G_1)\right|$$

We find the result after calculation by using table 1.

$$ABC_4(G_1)=\frac{\sqrt{15}}{6}(144)+\frac{\sqrt{14}}{6}(288)+\frac{\sqrt{3}}{3}(168)+\frac{\sqrt{11}}{6}(24)+\frac{\sqrt{10}}{6}(216)$$
$$+\frac{1}{2}(120)+\frac{\sqrt{78}}{18}(96)+\frac{\sqrt{210}}{30}(192)+\frac{\sqrt{30}}{12}(24)+\frac{\sqrt{5}}{5}(96)$$
$$+\frac{4}{9}(12)+\frac{\sqrt{170}}{30}(24)+\frac{2\sqrt{36}}{10}(36)$$

The index GA_5 can be calculated by using Geometric Arithmetic, then we find the result after calculation by using table 1.

$$GA_5(G_1)=\frac{4\sqrt{3}}{7}(144)+\frac{2\sqrt{2}}{3}(288)+\frac{2\sqrt{6}}{5}(168)+\frac{12}{13}(24)+1(216)$$
$$+\frac{4\sqrt{3}}{7}(120)+\frac{2\sqrt{6}}{5}(96)+\frac{\sqrt{15}}{4}(192)+\frac{12\sqrt{2}}{17}(24)+\frac{4\sqrt{5}}{9}(96)$$
$$+1(12)+\frac{6\sqrt{10}}{19}(24)+1(36)$$

□

Theorem 4.5: For the middle graph benzenoid planar octahedron network G_1, Hyper Zagreb index

then we have

$$HNI(G_1)(MBPOH) = 241608$$

Proof: Let can be calculated by using Hyper Zagreb index

$$\begin{aligned} HNI(G_1) = {} & 49|E_1(G_1)| + 81|E_2(G_1)| + 100|E_3(G_1)| + 169|E_4(G_1)| + 144|E_5(G_1)| + 196|E_6(G_1)| \\ & + 225|E_7(G_1)| + 256|E_8(G_1)| + 289|E_9(G_1)| + 324|E_{10}(G_1)| + 324|E_{11}(G_1)| \\ & + 361|E_{12}(G_1)| + 400|E_{13}(G_1)| \end{aligned}$$

$$\begin{aligned} HNI(G_1) = {} & 49(144) + 81(288) + 100(168) + 169(24) + 144(216) + 196(120) + 225(96) \\ & + 256(192) + 289(24) + 324(96) + 324(12) + 361(24) + 400(36) \end{aligned}$$

We find the result after calculation by using table 1

$$\Rightarrow HNI(G_1)(MBPOH) = 241608$$

$$\square$$

Theorem 4.6: Let G_1 *be* the middle graph benzenoid planar octahedron network MBPOH; its Harmonic indices then

$$\begin{aligned} H(G_1) = {} & \frac{2}{7}(144) + \frac{2}{9}(288) + \frac{2}{10}(168) + \frac{2}{13}(24) + \frac{2}{12}(216) + \frac{2}{14}(120) + \frac{2}{15}(96) \\ & + \frac{2}{16}(192) + \frac{2}{17}(24) + \frac{2}{18}(96) + \frac{2}{18}(12) + \frac{2}{19}(24) + \frac{2}{20}(36) \end{aligned}$$

Proof: Let G_1 *be* the middle graph benzenoid planar octahedron network. The outcome values find from table 1 by Harmonic index.

$$\begin{aligned} H(G_1) = {} & \frac{2}{7}|E_1(G_1)| + \frac{2}{9}|E_2(G_1)| + \frac{2}{10}|E_3(G_1)| + \frac{2}{13}|E_4(G_1)| + \frac{2}{12}|E_5(G_1)| + \frac{2}{14}|E_6(G_1)| + \frac{2}{15}|E_7(G_1)| \\ & + \frac{2}{16}|E_8(G_1)| + \frac{2}{17}|E_9(G_1)| + \frac{2}{18}|E_{10}(G_1)| + \frac{2}{18}|E_{11}(G_1)| + \frac{2}{19}|E_{12}(G_1)| + \frac{2}{20}|E_{13}(G_1)| \end{aligned}$$

We find the result after calculation by using table 1

$$H(G_1) = \frac{2}{7}(144) + \frac{2}{9}(288) + \frac{2}{10}(168) + \frac{2}{13}(24) + \frac{2}{12}(216) + \frac{2}{14}(120) + \frac{2}{15}(96)$$
$$+ \frac{2}{16}(192) + \frac{2}{17}(24) + \frac{2}{18}(96) + \frac{2}{18}(12) + \frac{2}{19}(24) + \frac{2}{20}(36)$$

□

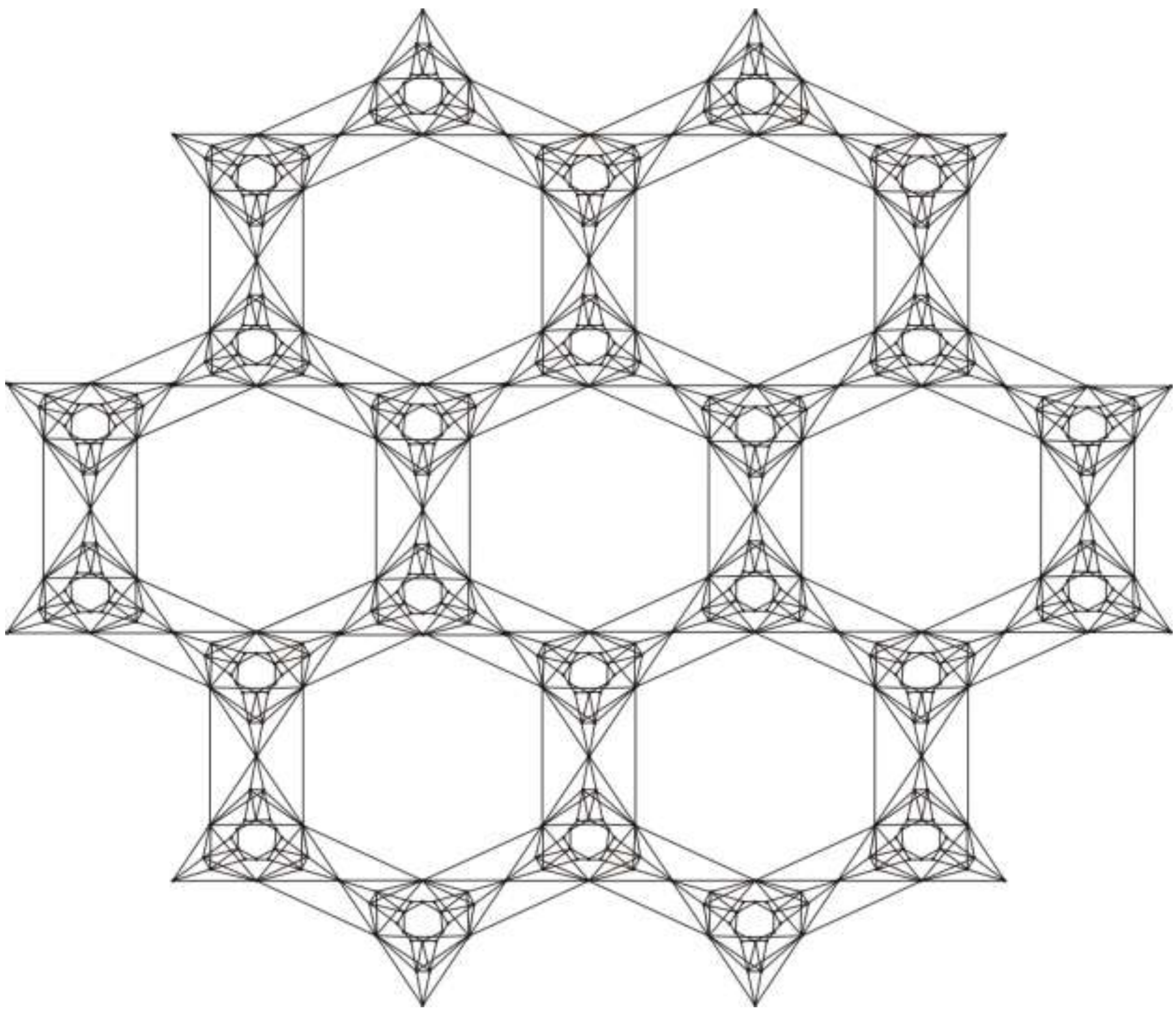


Fig 1. Middle Graph Benzenoid planar octahedron network MBDPOH

SECTION 2

Theorem 4.7: Let the benzenoid planar octahedron network BPOH(m); its first Zagreb index is equal to

$$N_1(BPOH(m)) = 873m^2 - 240m$$

Proof. Let G_1(m) be benzenoid planar octahedron network BPOH(m), shown in Fig.1, where m ≥ 2 and G_1(m) has 45m^2 – 3m vertices and the edges set of G_1 depending on the degree of the end vertices is divided into five parts.

Table 2. Degree-based edge partition for BPOH(m)

$\rho(u_i), \rho(v_j)$	Number of edges	$\rho(u_i), \rho(v_j)$	Number of edges
(3,3)	36m^2	(4,8)	12m
(3,4)	12m	(8,8)	18m^2-12m
(3,8)	36m^2-12m		

Also, we find the result by following the table 1 and using 1st Zagreb indices, we have

$$N_1(G_1(m)) = 6|E_1(G_1(m))| + 7|E_2(G_1(m))| + 11|E_3(G_1(m))| + 12|E_4(G_1(m))| + 16|E_5(G_1(m))|$$
$$= 6(36m^2) + 7(12m) + 11(36m^2 - 12m) + 12(12m) + 16(18m^2 - 12m)$$

we obtained the following result after calculation

$$\Rightarrow N_1(BPOH(m)) = 873m^2 - 240m$$

In this following theorem we use the 2nd Zagreb index

□

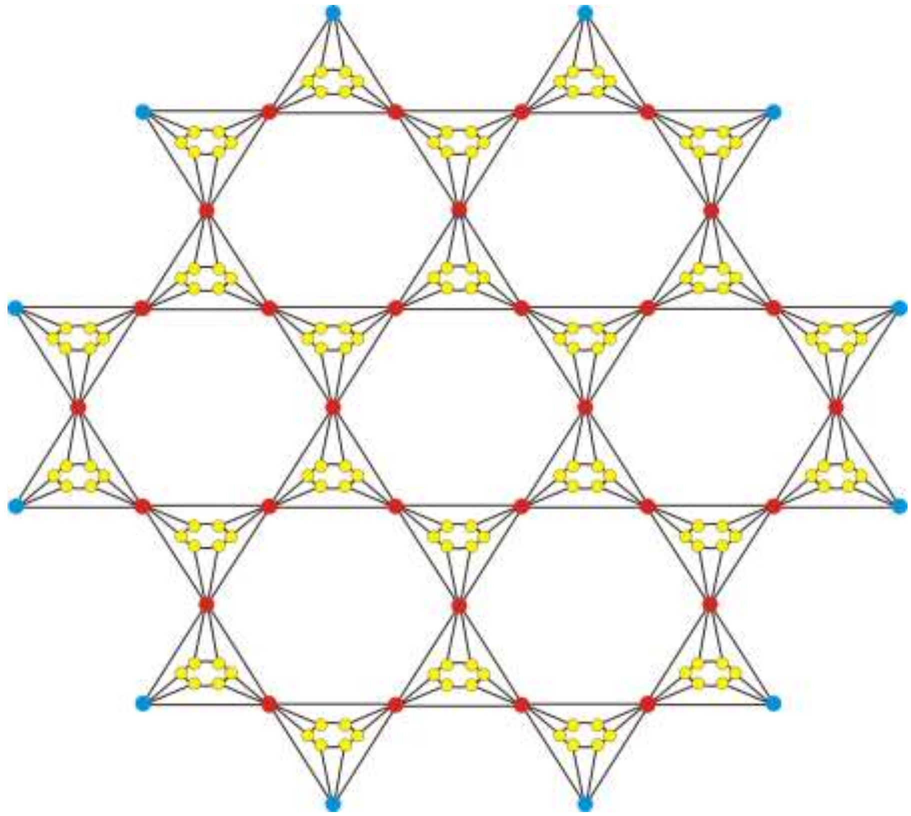


Fig. Benzenoid planar octahedron network BPOH (2)

Theorem 4.8: For the benzenoid planar octahedron network G_1(m), its 2nd Zagreb index is equal to

$$N_2(BPOH(m)) = 2340m^2 - 528m$$

Proof: Consider G_1(m) represent the benzenoid planar octahedron network. Then get find the result from table 1, use edge partition and use 2nd Zagreb indices, we have

$$N_2(G_1(m)) = 9|E_1(G_1(m))| + 12|E_2(G_1(m))| + 24|E_3(G_1(m))| + 32|E_4(G_1(m))| + 64|E_5(G_1(m))|$$

$$=9(36m^2) + 12(12m) + 24(36m^2 - 12m) + 32(12m) + 64(18m^2 - 12m)$$

we obtained the following result after calculation

$$\Rightarrow N_2(BPOH(m)) = 2340m^2 - 528m$$

□

Theorem 4.9: Let G_1(m) be the benzenoid planar octahedron network BPOH(m) $m \geq 2$, its Harmonic index is equal to

$$H(BPOH(m)) = \frac{7461}{308}m^2 - \frac{37}{22}m$$

Proof: Consider K_1(m) represent the benzenoid planar octahedron network BPOH(m), With harmonic index us and the outcome produced by the edge partition, we have

$$H(G_1(m)) = \frac{2}{6}|E_1(G_1(m))| + \frac{2}{7}|E_2(G_1(m))| + \frac{2}{11}|E_3(G_1(m))| + \frac{2}{12}|E_4(G_1(m))| + \frac{2}{16}|E_5(G_1(m))|$$

$$= \frac{2}{6}(36m^2) + \frac{2}{7}(12m) + \frac{2}{11}(36m^2 - 12m) + \frac{2}{12}(12m) + \frac{2}{16}(18m^2 - 12m)$$

we obtained the following result after calculation

$$\Rightarrow H(BPOH(m)) = \frac{7461}{308}m^2 - \frac{37}{22}m$$

Theorem 4.10: Let G_1(m) be the benzenoid planar octahedron network MBPOH(m), its Hyper Zagreb index is equal to

$$HN(BPOH(m)) = 10260m^2 - 2208m$$

Proof: Consider G_1(m) represent the benzenoid planar octahedron network BPOH(m) With hyper zagreb indices use and the outcome produced by the edge partition, we have

$$HN(G_1(m)) = 36|E_1(G_1(m))| + 49|E_2(G_1(m))| + 121|E_3(G_1(m))| + 144|E_4(G_1(m))| + 256|E_5(G_1(m))|$$

$$= 36(36m^2) + 49(12m) + 121(36m^2 - 12m) + 144(12m) + 256(18m^2 - 12m)$$

we obtained the following result after calculation

$$\Rightarrow HN(BPOH(m)) = 10260m^2 - 2208m$$

□

Theorem 4.11: Let G_1(m) be the benzenoid planar octahedron network BPOH(m), its Sum connectivity index is equal to

$$SCI(G_1(m)) = \frac{30051}{1000}m^2 - \frac{1381}{1000}m$$

Proof: Consider G_1(m) represent the middle graph benzenoid planar octahedron network BPOH(m), With sum connectivity index use and the outcome produced table 1 by the edge partition, we have

$$SCI(G_1(m)) = \frac{\sqrt{6}}{6}\left|E_1(G_1(m))\right| + \frac{\sqrt{7}}{7}\left|E_2(G_1(m))\right| + \frac{\sqrt{11}}{11}\left|E_3(G_1(m))\right| + \frac{\sqrt{3}}{6}\left|E_4(G_1(m))\right| + \frac{1}{4}\left|E_5(G_1(m))\right|$$
$$= \frac{\sqrt{6}}{6}(36m^2) + \frac{\sqrt{7}}{7}(12m) + \frac{\sqrt{11}}{11}(36m^2 - 12m) + \frac{\sqrt{3}}{6}(12m) + \frac{1}{4}(18m^2 - 12m)$$

we obtained the following result after calculation

$$\Rightarrow SCI(G_1(m)) = \frac{30051}{1000}m^2 - \frac{1381}{1000}m$$

Table 3. Degree based edge partition for BPOH(m)

$\rho(u_i), \rho(v_j)$	Number of edges	$\rho(u_i), \rho(v_j)$	Number of edges
(3,3)	$108m^2$-132m+48	(4,8)	24m-12
(3,4)	12m-12	(8,8)	$54m^2$-162m+84
(3,8)	$108m^2$-132m+48		

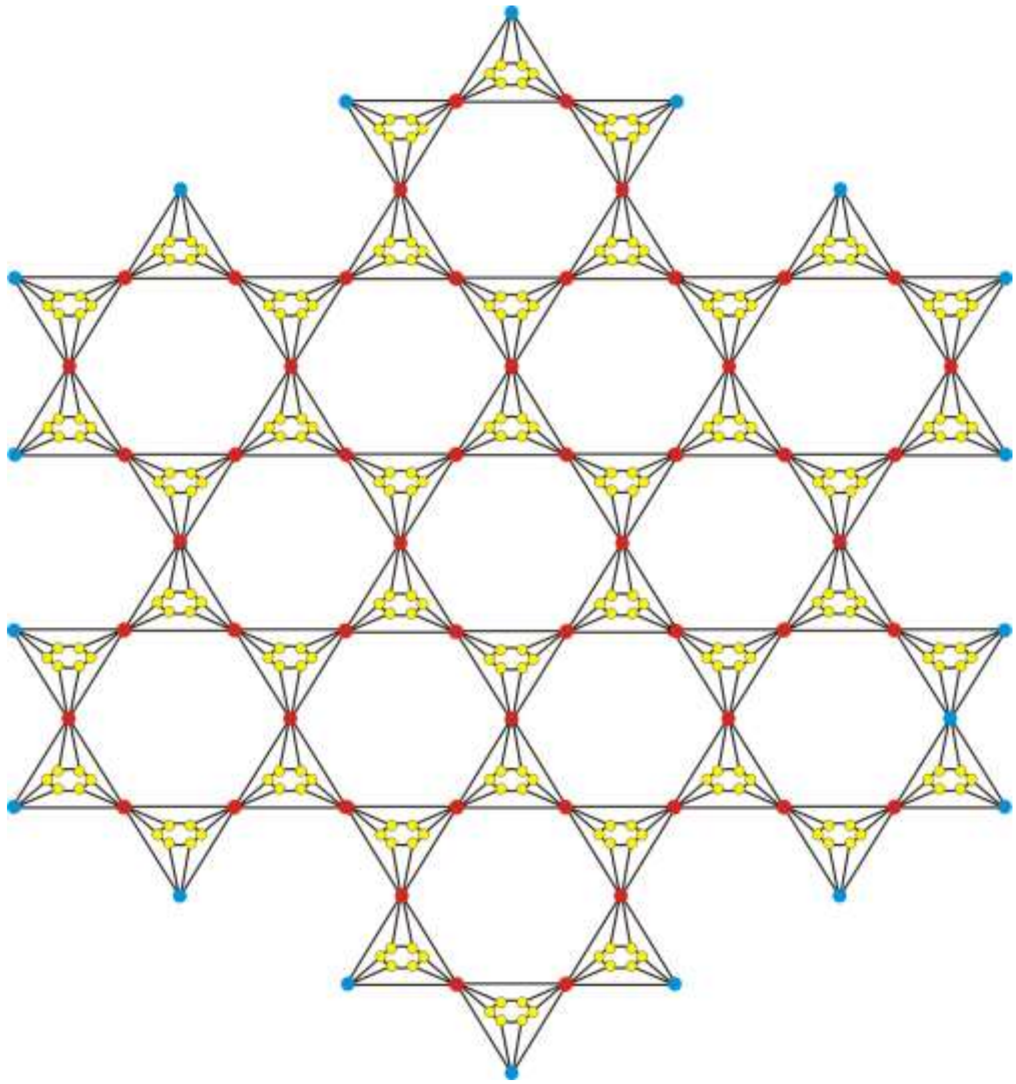


Fig. Benzenoid-dominating planar octahedron network BDPOH (2)

The following theorem implies that we evaluate Harmonic index of benzenoid planar octahedron network BPOH(m)

Theorem 4.12: Let G_2(m) be the benzenoid planar octahedron network MBPOH(m), its Harmonic index is equal to

$$H(BPOH(m)) = \frac{2745}{44}m^2 - \frac{2391}{28}m + \frac{4589}{154}$$

Proof: Let G_2 (m) be the benzenoid-dominating planar octahedron network BPOH(m), Table 2's results can be obtained by utilizing its Harmonic index, which equals

$$H(G_1(m)) = \frac{2}{6}|E_1(G_1(m))| + \frac{2}{7}|E_2(G_1(m))| + \frac{2}{11}|E_3(G_1(m))| + \frac{2}{12}|E_4(G_1(m))| + \frac{2}{16}|E_5(G_1(m))|$$

$$= \frac{2}{6}(108m^2 - 132m + 48) + \frac{2}{7}(24m - 12) + \frac{2}{11}(108m^2 - 132m + 48) + \frac{2}{12}(24m - 12) + \frac{2}{16}(54m^2 - 162m + 84)$$

we obtained the following result after calculation

$$\Rightarrow H(BPOH(m)) = \frac{2745}{44}m^2 - \frac{2391}{28}m + \frac{4589}{154}$$

$\square$

Theorem 4.13: Let G_2(m) be the benzenoid planar octahedron network BPOH(m), its Inverse sum index is equal to

$$ISI(BPOH(m)) = \frac{6750}{11}m^2 - \frac{7346}{7}m + \frac{35432}{77}$$

Proof: Let can be calculated by using Inverse sum index

$$ISI(G_1(m)) = \frac{9}{6}|E_1(G_1(m))| + \frac{12}{7}|E_2(G_1(m))| + \frac{24}{11}|E_3(G_1(m))| + \frac{32}{12}|E_4(G_1(m))| + \frac{64}{16}|E_5(G_1(m))|$$

$$\mathrm{ISI}(\mathrm{G}_1(\mathrm{m})) = \frac{9}{6}(108m^2 - 132m + 48) + \frac{21}{7}(24m - 12) + \frac{24}{11}(108m^2 - 132m + 48) + \frac{32}{12}(24m - 12) + \frac{64}{16}(54m^2 - 162m + 84)$$

We find the result after calculation by using table 2

$$\Rightarrow ISI(BPOH(m)) = \frac{6750}{11}m^2 - \frac{7346}{7}m + \frac{35432}{77}$$

CONCLUSION

In this thesis, we discussed degree-based topological indices in our thesis. We first draw middle graph benzenoid planar octahedron networks, then draw dominating-middle graph benzenoid planar octahedron networks, and calculate topology indices based on the degree of the middle graph benzenoid planar octahedron networks to determine the degree of each vertex of the graph we discuss a number of topological indices, including the Harmonic, ABC (Atom Bond Connectivity), Geometric, Randic, and Forgotten Topological indices. We covered the creation of molecular graphs and middle graph benzenoid planar octahedron networks, and we have some fresh results employing topological indices.

Result for Middle Graph of Benzenoid Planar Octahedron Network.

$$\bullet \mathrm{R}(G_1)\ \textit{Randic index} = \frac{1}{\sqrt{12}}(144) + \frac{1}{\sqrt{18}}(288) + \frac{1}{\sqrt{24}}(168) + \frac{1}{\sqrt{36}}(24) + \frac{1}{\sqrt{36}}(216) + \frac{1}{\sqrt{48}}(120) + \frac{1}{\sqrt{54}}(96) + \frac{1}{\sqrt{60}}(192) + \frac{1}{\sqrt{72}}(24) + \frac{1}{\sqrt{80}}(196) + \frac{1}{\sqrt{81}}(12) + \frac{1}{\sqrt{90}}(24) + \frac{1}{\sqrt{100}}(36)$$

$$\bullet H(G_1)\ \textit{Harmonic index} = \frac{2}{7}(144) + \frac{2}{9}(288) + \frac{2}{10}(168) + \frac{2}{13}(24) + \frac{2}{12}(216) + \frac{2}{14}(120) + \frac{2}{15}(96) + \frac{2}{16}(192) + \frac{2}{17}(24) + \frac{2}{18}(96) + \frac{2}{18}(12) + \frac{2}{19}(24) + \frac{2}{20}(36)$$

$$\bullet N_1(BPOH(m))\ 1^{st}\ \textit{Zagreb index} = 873m^2 - 240m$$

$$\bullet N_2(BPOH(m))\ 2^{nd}\ \textit{Zagreb index} = 2340m^2 - 528m$$

$$\bullet HNI(G_1)(MBPOH)\ \textit{Hyper Zagrebindex} = 241608$$

$$\bullet SCI(G_1(m))\ \textit{Sum connectivity index} = \frac{30051}{1000}m^2 - \frac{1381}{1000}m$$

$$\bullet F(MBPOH)\ \textit{forgotten index} = 118268$$

$$\bullet \operatorname{Re} ZG_1(G_1)\ \text{1st}\ \textit{reclassified Zagreb indices} = \frac{1070136}{455} + \frac{420218}{255} + \frac{5530}{19}$$

$$\bullet \operatorname{Re} ZG_2(G_1)\ \text{2nd}\ \textit{reclassified Zagreb indices} = \frac{4737}{5}$$

$$\bullet \operatorname{Re} ZG_3(G_1)\ \text{3rd}\ \textit{reclassified Zagreb indices} = 844488$$

$$\begin{aligned}\bullet GA_5(G_1)\ \textit{Geometric Arithmetic} = {} & \frac{4\sqrt{3}}{7}(144) + \frac{2\sqrt{2}}{3}(288) + \frac{2\sqrt{6}}{5}(168) + \frac{12}{13}(24) + 1(216) \\ & + \frac{4\sqrt{3}}{7}(120) + \frac{2\sqrt{6}}{5}(96) + \frac{\sqrt{15}}{4}(192) + \frac{12\sqrt{2}}{17}(24) + \frac{4\sqrt{5}}{9}(96) \\ & + 1(12) + \frac{6\sqrt{10}}{19}(24) + 1(36)\end{aligned}$$

$$\bullet ISI(BPOH(m))\ \textit{Inverse sum index} = \frac{6750}{11}m^2 - \frac{7346}{7}m + \frac{35432}{77}$$

$$\begin{aligned}\bullet ABC_4(G_1)\ \textit{atom} - \textit{bond connectivity} = {} & \frac{\sqrt{15}}{6}(144) + \frac{\sqrt{14}}{6}(288) + \frac{\sqrt{3}}{3}(168) + \frac{\sqrt{11}}{6}(24) + \frac{\sqrt{10}}{6}(216) \\ & + \frac{1}{2}(120) + \frac{\sqrt{78}}{18}(96) + \frac{\sqrt{210}}{30}(192) + \frac{\sqrt{30}}{12}(24) + \frac{\sqrt{5}}{5}(96) \\ & + \frac{4}{9}(12) + \frac{\sqrt{170}}{30}(24) + \frac{2\sqrt{36}}{10}(36)\end{aligned}$$

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